

Konstantinos Kanellopoulos

Electrical & Computer Engineering Student | Research Assistant

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PROFILE

Electrical and Computer Engineering student and **Research Assistant in the VVR group** (University of Patras) under Prof. Konstantinos Moustakas, with strong interests in **simulation, computer graphics, digital twins, neuromorphic computing, robotics, and AI**. Experienced in building complete **hardware and software systems**: from physically based real-time renderers and traffic/flood simulation pipelines to robot control, event-driven backends, and cryptographic analysis.

Hands-on with **Python, C++, Java, Dart, ROS2, VTK/PyVista, NumPy/SciPy, Linux, CAD systems, and modern software engineering workflows** (regression testing, profiling, reproducible pipelines).

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, Dart, C, SQL, Bash, JavaScript

Graphics & Geospatial: VTK / PyVista, Physically Based Rendering (PBR), Image-Based Lighting, Post-processing (SSAO, FXAA), GPU Instancing, Procedural Modeling, GIS (OpenStreetMap, Overture Maps, DEM/DTM/nDSM rasters, UTM/EGM2008), Photogrammetry Data, Computational Geometry

Simulation & Modeling: Digital Twins, Microscopic Traffic Simulation (IDM, SUMO), Hydrodynamic Flood Modeling, Agent-Based Simulation, Air-Quality and Noise Models, Wildfire Spread (Rothermel)

AI & Reinforcement Learning: Reinforcement Learning (PPO), Spiking Neural Networks (ALIF, R-STDP, e-prop), Imitation Learning (DAgger), Statistical Evaluation, Simulation-Based Training, Control Systems, Physics Simulations

Robotics: ROS2, MuJoCo, Kinematics, LIPM Walking Control, Motion Planning, Robotic Arm Control, Sensor/Actuator Integration

Software Engineering: Git, Linux, Event-Driven Architecture, REST APIs, Object-Oriented Design, Automated Testing (regression-first), Performance Profiling & Optimization, Real-Time Systems

Engineering & Hardware: Fusion 360, CAD Design, 3D Printing, Embedded Systems Fundamentals, Optics, Hardware Prototyping

Cybersecurity: Reverse Engineering, Cryptography, XOR Ciphers, Base64 Analysis, Known-Plaintext Attacks

EDUCATION

University of Patras

Department of Electrical and Computer Engineering

2022 – 2027 (Expected)

Patras, Greece

- Current GPA: **7.45/10**

Anavryta Model High School

High School Diploma – Grade: 19.7/20

2019 – 2022

Athens, Greece

PROFESSIONAL EXPERIENCE

VVR Group, University of Patras

Research Assistant – supervised by Prof. Konstantinos Moustakas

2026 – Present

Patras, Greece

- Developing a **photoreal digital twin of Beirut** that couples real-time traffic and **flood** simulation with drone-survey data (see Technical Projects).

Poradix.gr

Junior Software Developer

Dec 2025 – May 2026

Remote

- Developed an **Event Store Manager** for handling customer communication and financial records.
- Designed and implemented **event-driven backend architecture** for real-time transaction tracking.

Othisi Educational Center

Exam Supervisor – ran examinations in strict compliance with regulations

2023 – 2025

TECHNICAL PROJECTS

Neuromorphic Control for Quadruped Robots

Undergraduate Thesis – AI-Hub 2026 Competition Submission

2026

[GitHub](#)

- Benchmarked **spiking neural networks** (ALIF, R-STDP, e-prop) against deep networks for LiDAR navigation of a **Unitree Go2** quadruped in MuJoCo under distribution shift; designed an **FPGA co-processor** as a memristor-crossbar analog.
- SNN **10.4×** cheaper in synaptic operations; to our knowledge the **first walking Go2 controlled by a spiking network**; statistically rigorous protocol (10 seeds, Holm–Bonferroni).

Digital Twin City & Flood Simulation (Beirut)

Python, VTK, Computational Geometry, GIS, PBR Rendering

[GitHub](#)

- Real-time city **digital twin** with microscopic traffic (IDM/SUMO), pedestrians, and public transport, plus safety, air-quality, and noise analyses.
- Integrated **flood simulation** results and a 5 cm **drone survey** (terrain, orthophoto, building heights) into a **physically based renderer**; 4× faster frames through GPU instancing.

Optical Engineering & Hardware Design

CAD, 3D Printing, Optics

- Designed and constructed a custom **Hadley/Newtonian reflecting telescope** (1 m focal length, 20 cm primary mirror) using **CAD modeling, 3D printing**, and precision optical alignment.

Security Research & Reverse Engineering

Python, Cryptography

- Reverse engineered and decrypted a proprietary mobile game database using **XOR cipher analysis, Base64 decoding**, and **known-plaintext attacks**.

Bipedal Robot Design & Robotic Arm Control

Fusion 360, MuJoCo, ROS2, C++

- Designed a custom **bipedal robot** in Fusion 360 and simulated it in MuJoCo with **standing stabilization** and **walking gait control** via the **Linear Inverted Pendulum Model (LIPM)**.
- Integrated sensors and actuators through **ROS2**.
- Developed robotic arm trajectory algorithms: [GitHub](#)

Other Software Projects

C++, AI, Simulation

- Multiplayer RTS game with custom graphics libraries ([GitHub](#)); “Enigma”, a Quizdom-style multiplayer platform.
- Wildfire prediction system based on the **Rothermel fire spread model**.

ACHIEVEMENTS & CERTIFICATIONS

- **EBEC Engineering Competition – University of Patras: 2nd place** for a real-time bus tracking platform and dashboard prototype.
- **1st Place, Sumo Robot Competition** of the **Robotics Club of the University of Patras** (autonomous robot design, build, and programming).
- **Hellenic Mathematical Society:** Qualified in the “Thales” and “Euclid” national competitions.
- **SAT Score:** 1420
- **Languages:** English (C2 Cambridge Proficiency), German (B2 Goethe)

VOLUNTEERING

ARCHELON – Sea Turtle Protection Society of Greece

Environmental Volunteer

3 Years

- Participated in field operations for the protection of *Caretta caretta* nesting sites and eggs.